sales data analytics in machine learning

```
from google.colab import files
uploaded = files.upload()
```

Choose files sales_data.csv

**sales_data.csv**(text/csv) - 1446012 bytes, last modified: 10/10/2026 - 100% done

```
Saving sales_data.csv to sales_data.csv
```

Double-click (or enter) to edit

```
import pandas as pd
df = pd.read_csv("sales_data.csv")
df
```

| tomer_Name | Product | Category | Reg |
|---|---|---|---|
| ıstomer_140 | Mouse | Accessories | Tirune |
| ıstomer_973 | Office Desk | Furniture | Coimba |
| stomer_1045 | Smartphone | Mobiles | Mad |
| stomer_1173 | Headphones | Accessories | Mad |
| stomer_1245 | Monitor | Electronics | Che |
| ... | ... | ... | |
| ıstomer_320 | Printer | Office Equipment | Sa |
| stomer_1345 | Desk Chair | Furniture | Tirune |
| stomer_1032 | Office Desk | Furniture | Tı |
| stomer_1076 | Mouse | Accessories | Mad |
| stomer_1024 | Monitor | Electronics | Coimba |

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_squared_error

x = df['Quantity'].values.reshape(-1,1)
y = df['Unit_Price'].values.reshape(-1,1)

x_train, x_test, y_train, y_test = train_test_split(x,y,test_size=0.2, random_state=42
model = LinearRegression()
model.fit(x_train, y_train)
y_pred = model.predict(x_test)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)
```

```
print(f"RMSE: {rmse_val}")
print(f"r2_score: {r2_val}")
print(f"Intercept: {model.intercept_[0]}")
print(f"Coefficient: {model.coef_[0][0]}")
```

```
RMSE: 14185.984341205072
r2_score: -0.00042096269454927615
Intercept: 12593.426233439115
Coefficient: -6.829640887132482
```